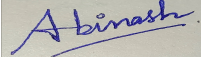
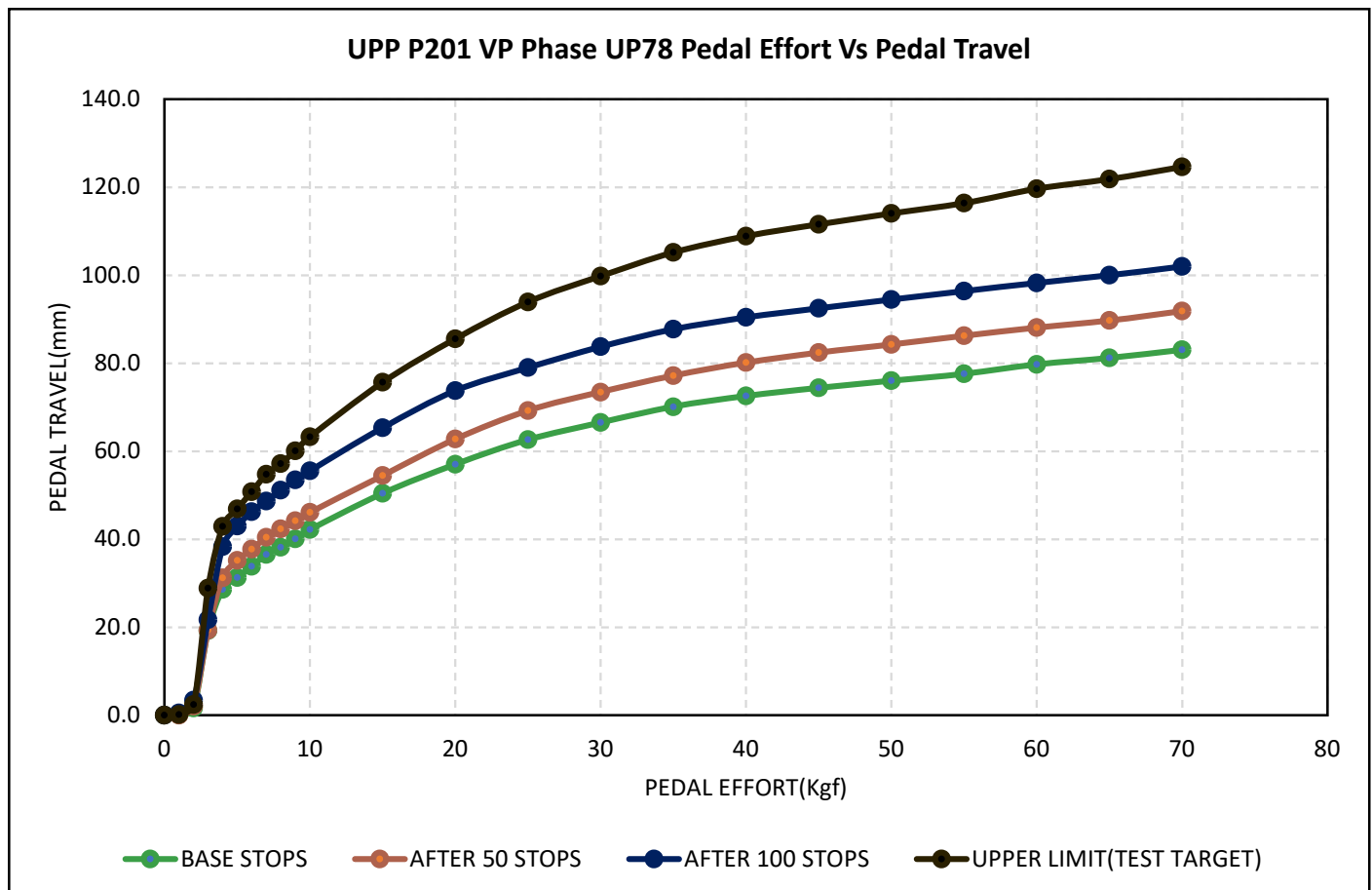
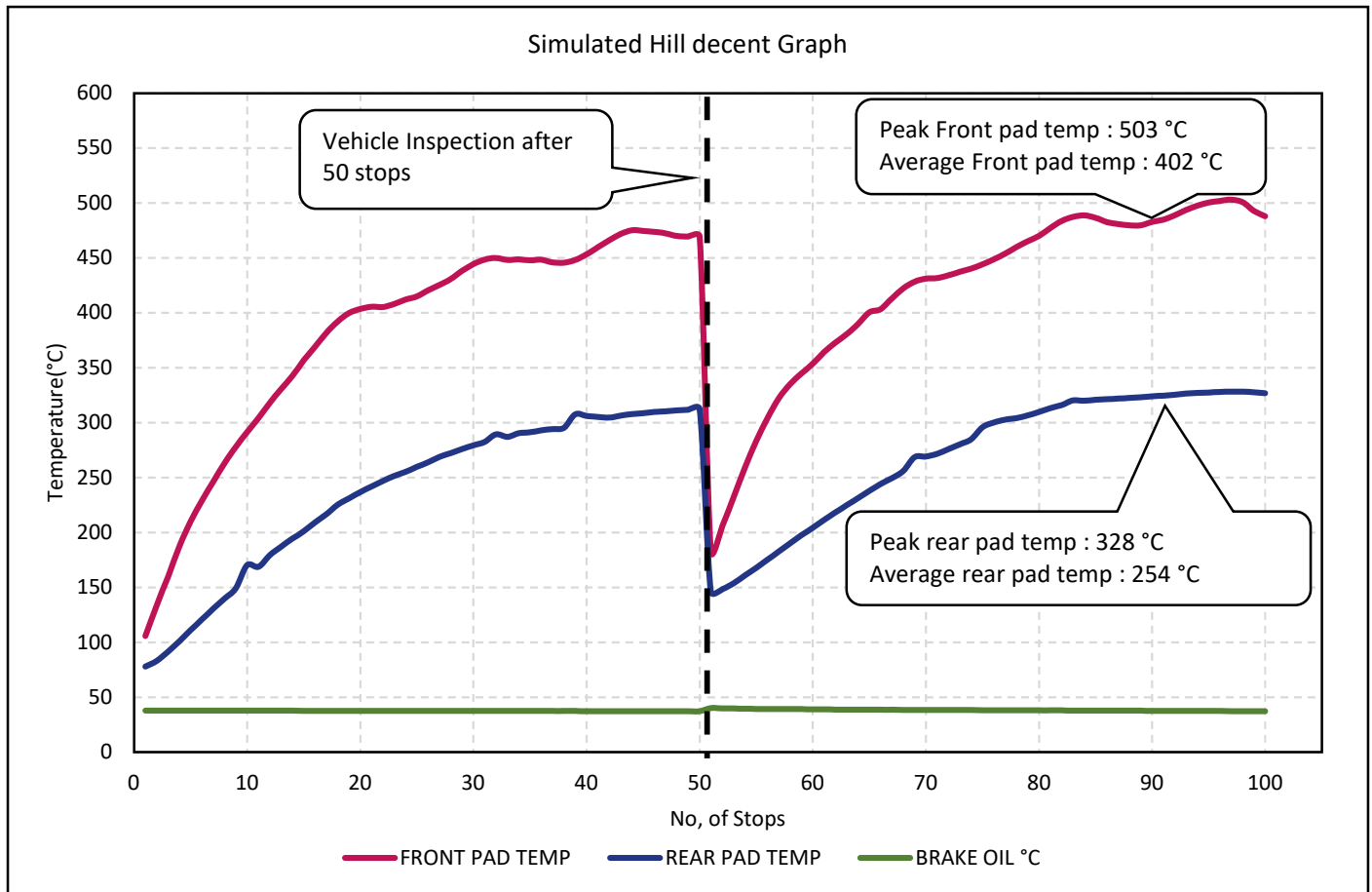
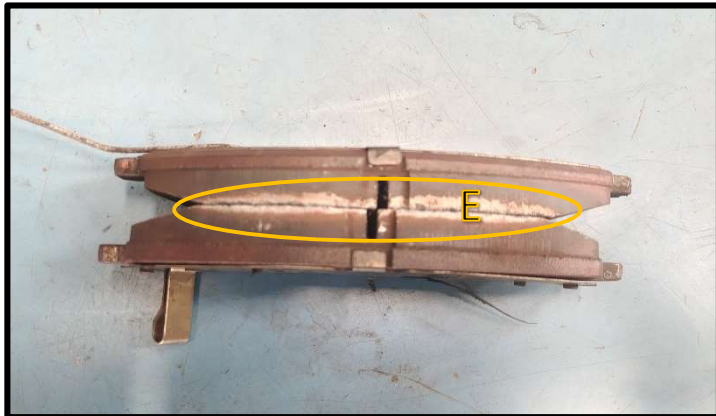
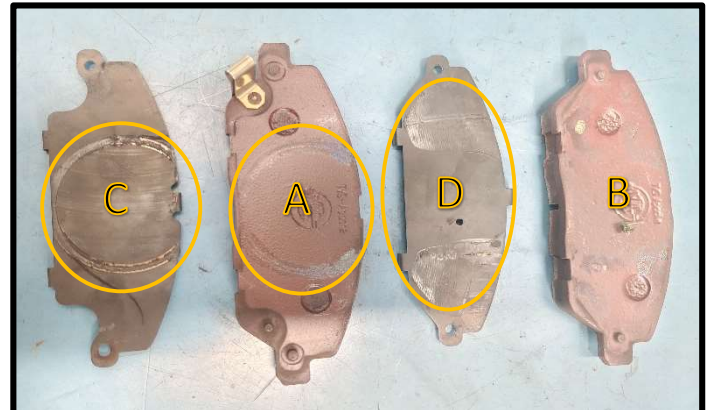
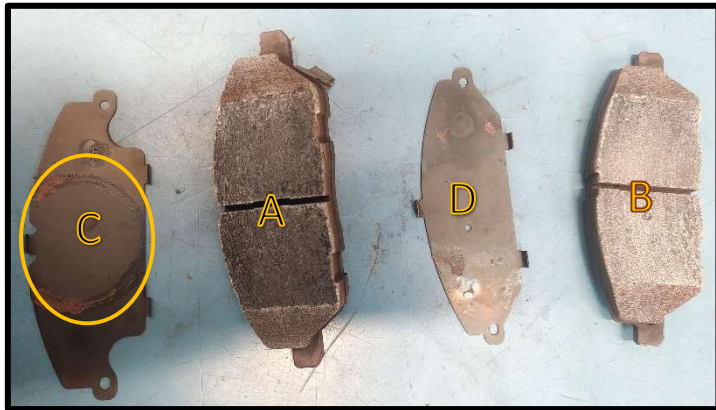


TPDS		TEST REPORT				
SIMULATED HILL DECENT TEST FOR BRAKE SYSTEM ON VP PHASE UP78 VEHICLE IN GVW CONDITION						
To : NAGRIKAR RAVI/MISHRA KRISHNA		Report No: TPDS/VT/RDP201/VP/2024/28051				
Project No: RDP201-UPP Goods-G2-Domestic		Report Date: 31-May-2024				
Objective:		To evaluate simulated hill decent test on P201 VP phase vehicle with respect to test target.				
Scope:		To conduct simulated hill decent test on P201 VP phase vehicle in GVW condition				
Vehicle Configuration:						
Foundation Brakes	BIL	Booster Size	9"			
Front Brake	C60, Single pot	Booster Ratio	5			
Front Disc Size, Eff radius	277 x 24, 116	Jump in	12 +/-3			
Front Brake Pad Grade	Apache 5409M	Pedal Ratio	3.5:1			
Rear Brake	25.4 mm (WC)	TMC Size	23.81			
Rear Drum size, Eff. radius	279 X 50.8, 279	Stroke-Primary/Secondary	17.5/18.5mm			
Rear Brake Pad Grade	RD0014	System Split	"X" Split			
Test:		06-C-06-0311-001-SIMULATED HILL DECENT TEST FOR BRAKE SYSTEM				
Observation:		Refer the table below for test observations:-				
Test Target		Observation			Remarks	
1. No abnormal/permanent black dust formation should happen on the wheel rim surface after the test. 2. At the completion of test, no evidence of crack, chip off, peel off, abnormal ash formation, scoring & glazing should happen on brake components such as brake disc/drums, friction material & brake components. 3. No Blistering, discoloration, peel-off & cap drop out should occur on wheel rim. 4. No discoloration of drum or disc should occur. 5. Friction material chip-off, pitting & material intrusion / interchange should not occur 6. No abnormal noise should occur during brake application. 7. No smoke or melting of wheel speed sensor or any other components should occur. 8. Change in disc mounting bolt & wheel nut torque should not be more than 25%. 9. No brake & ABS malfunction lamp should glow on the instrument cluster. 10. Pedal travel increase should not be more than 50% of the base reading (upper limit) taken before the start of test 11. At the end of the two set of 50 deceleration cycles, the brakes system should remain in a serviceable condition.		1. Peak temperatures recorded are 502 °C and 328 °C for front pad and rear pad respectively. 2. No abnormal/permanent black dust formation observed on the wheel rim surface. 3. No evidence of crack, chip off, peel off observed on brake components such as brake disc/drums, friction material & brake components. 4. Abnormal ash formation, scoring & glazing observed on brake components such as brake disc/drums, friction material & brake components. 5. Blistering, discoloration, peel-off & cap drop out are not observed on wheel rim. 6. Discoloration observed from disc and drum. 7. Friction material intrusion / interchange are present. 8. During braking spark started from 17th stop and Red Hot on disc started from 30th stop. 9. Squeal noise observed (52 to 63 stops) from rear at the end stop. 10. No smoke or melting of wheel speed sensor or any other components observed 11. Change in disc mounting bolt & wheel nut torque is less than 25%. 12. No brake malfunction lamp indication in the instrument cluster. 13. Pedal travel increase is less than 50% of the base reading taken before the start of test 14. At the end of the two set of 50 deceleration cycles, the brakes system is in serviceable condition.			Test vehicle is meeting the test requirement	
Conclusion:		Simulated hill descent test on P201 VP phase vehicle is meeting the test target.				
Recommendation:		Need attention to point no 9. Squeal noise observed (52 to 63 stops) from rear at the end stop. Concern raised in CRRMS(CRRMS no: RDP201-INC-00944)				
Attachments:		Data, graph				
Test Conducted / Reported by  Abinash Abinash		Reviewed by  MITTAL SAPAN		Approved by  Rathakrishnan M		
CC		SHABIR KARNURE/LALASURE SANTOSH				

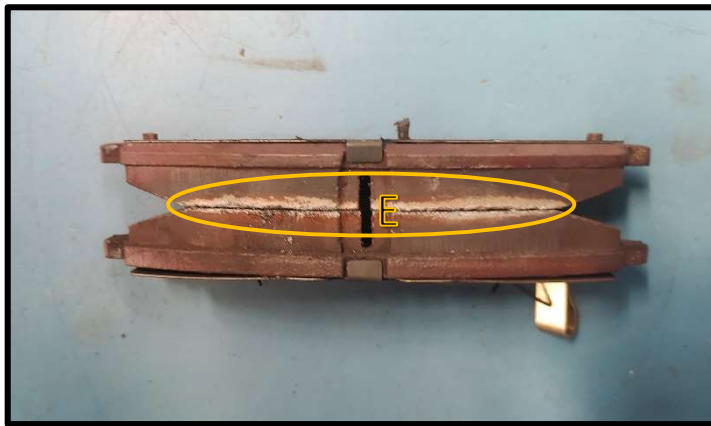
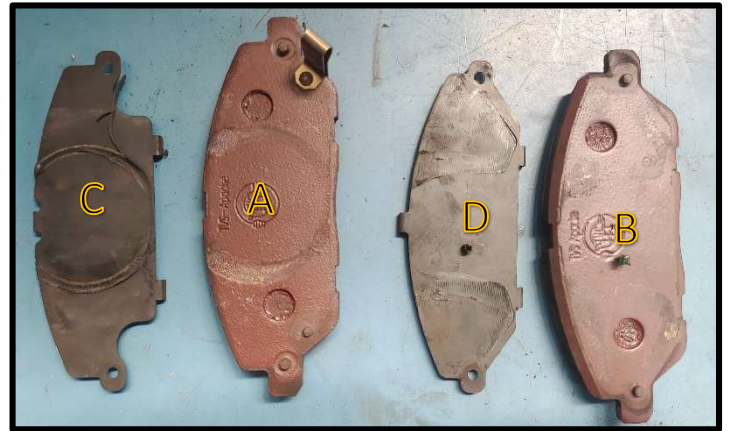
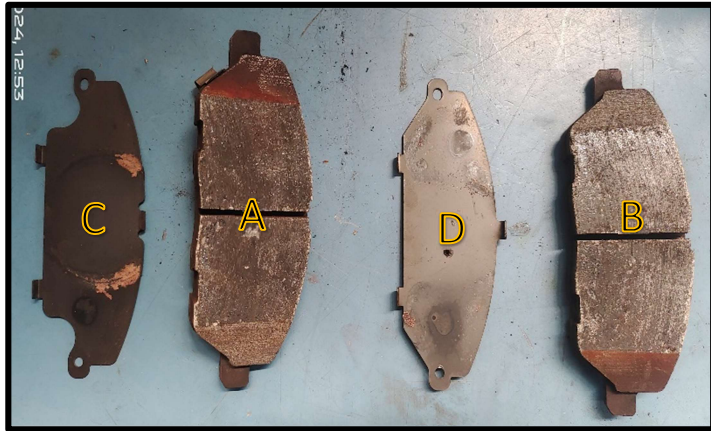
TPDS		VEHICLE CONFIGURATION			
Project No : RDP201-UPP Goods-G2-Domestic			Report No : TPDS/VT/RDP201/VP/2024/28051		
Vehicle Details					
Parameter		UPP G2 3D15 2550WB Diesel FSD			
Model		Diesel 3D15 AC Configuration			
Reg No / S No		R6B15023			
Engine Model & No		1.5L, 3 cylinder BS6.2 MT CUR6B44576			
C.G.Height(MM)		902 MM			
Transmission /Gearbox		5 Speed Manual Transmission			
Drive		2WD			
Axle Ratio		4.55			
Wheel / Tyre Details		Tubeless 195 R15 37 PSI and 58 PSI MRF			
Front Suspension		Semi elliptical leaf spring With			
Rear Suspension		Semi elliptical leaf spring With			
Steering Details		Youngshin/Ranes, Vane pump			
Speedo Make/Raio		RCBT			
Odometer		--			
Wheel Track(mm)		FRONT - 1465, REAR - 1375			
Wheel Base(mm)		2550			
F/R Ratio		--			
Dyn.Rolling Radius		--			
Weight Transfer Ratio		--			
Weight Distribution:					
UNLADEN	Curb Wt (kg) FAW (kg) RAW (kg) ETW (kg)	1631 1157 474 1631			
LADEN	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	3038 1400 1638 1260			
OVERLOAD	GVW (kg) FAW (kg) RAW (kg) Pay Load (kg)	3800 2305 1495 3800			
Amb Temperature		35 deg c			
Wind Velocity		<3 m/s			
Gust		< 5 m/s			
Rel. Humidity		--			
Weather		Sunny			
Road Surface		TAR			
Instruments Used		LINK LOGGER & L350			
Test Site		MSPT			
Test Driver		Abinash			
Test Engineer		Abinash			
Test Date		07-05-2024			



FLH BRAKE PAD AND DISC



- A - Inner Pad: Glazing, Decolorization and bluish color observed.
- B - Outer Pad: Glazing, Decolorization observed.
- C - Inner pad Sim: Piston marks observed.
- D - Outer pad Sim: Caliper marks observed.
- E - Ash formation found on front brake Pad.
- F - Disc: Scoring marks, Glazing, decolorization observed.



A - Inner Pad: Glazing, Decolorization and bluish color observed.

B - Outer Pad: Glazing, Decolorization observed.

C- Inner pad Sim: Piston marks observed.

D - Outer pad Sim: Caliper marks observed.

E - Ash formation found on front brake Pad.

F - Disc: Scoring marks, Glazing, decolorization observed.

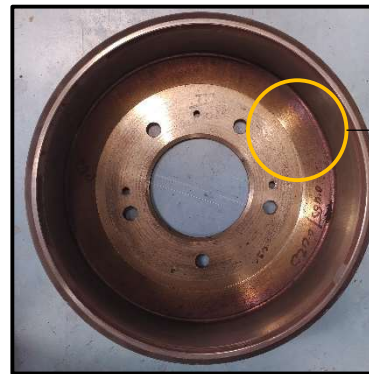
RLH LINER AND DRUM



LEADING SHOE



TRAILING SHOE



Bluish on drum



DRUM

RRH LINER AND DRUM



LEADING SHOE



TRAILING SHOE



Bluish on drum



DRUM